

The Motorola Pager

How a 1980s pager crisis gave birth to Six Sigma and its famous defect target.

The Crisis

In the early 1980s Motorola was a giant in electronics, but its pager division was losing market share to Japanese competitors. The Japanese pagers lasted longer, broke down less, and cost less to make. When executives investigated, they found the rivals were not just inspecting at the end of the line. They built processes so precise that defects barely happened in the first place.

Bill Smith's Target

In 1986 a Motorola engineer named Bill Smith proposed a radical quality goal of no more than 3.4 defects per million opportunities. That target is where the name Six Sigma comes from. Sigma is a Greek letter statisticians use to measure how spread out or inconsistent a process is, and a Six Sigma process is so tightly controlled that almost nothing goes wrong.

Why It Matters

By tracking and fixing root causes with data, Motorola saved billions over the following years. Companies like General Electric and Honeywell adopted the same discipline, proving extreme consistency works in any industry. While Lean targets waste, Six Sigma targets variation.

Remember this

Six Sigma means 3.4 defects per million, built by preventing defects rather than inspecting for them.